# Osptek Display

# SPECIFICATION FOR 1.54''EPD

Model No:

**EPD0154A281A0**

| Version | Content | Date | Producer |
|---|---|---|---|
| A0 | New release | 2026/1/15 | Eddie Huang |
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## 1. General Description

EPD0154A281A0 is an Active Matrix Electrophoretic Display (AMEPD), with interface and a reference system design. The 1.54” active area contains 200*200 pixels, and has B/W full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

## 2. Features

- 200*200 pixels display
- High contrast
  High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Low voltage detect for supply voltage
- High voltage ready detect for driving voltage
- Internal temperature sensor
- Waveform stored in On-chip OTP
- 10-byte OTP space for module identification
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- I2C signal master interface to read external temperature sensor/ built-in temperature sensor

## 3. Application

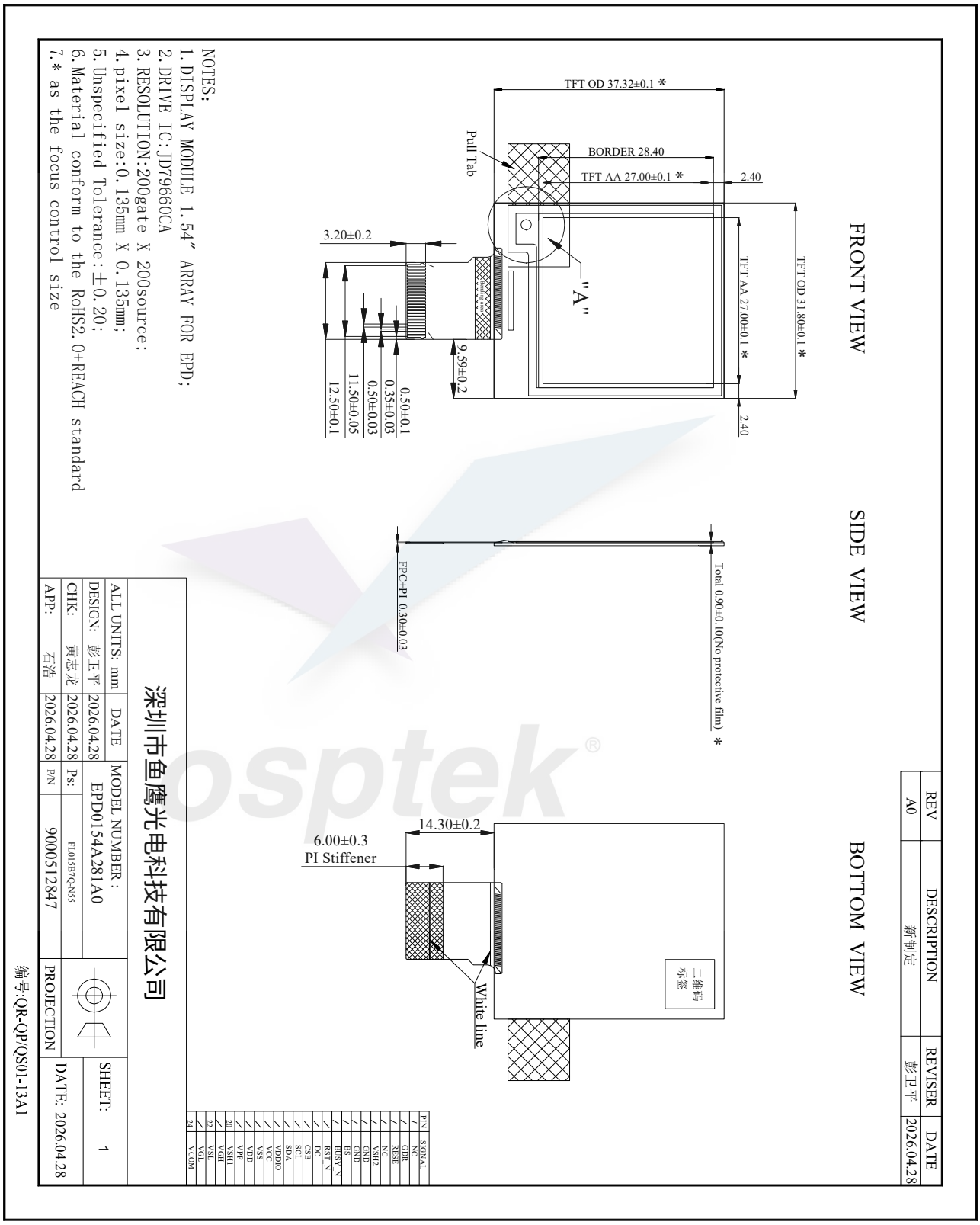
Electronic Shelf Label System

## 4. Mechanical Specification

| Parameter | Specifications | Unit | Remark |
|---|---|---|---|
| Screen Size | 1.54 | Inch | |
| Display Resolution | 200(H)×200(V) | Pixel | Dpi:188 |
| Active Area | 27.00 (H)×27.00 (V) | mm | |
| Pixel Pitch | 0.135×0.135 | mm | |
| Pixel Configuration | Square | | |
| Outline Dimension | 37.32(H)×31.80(V) ×0.86(D) | mm | Without masking film |
| Weight | 2.1±0.2 | g | |

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# 5. Mechanical Drawing of EPD module




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## 6. Input/output Terminals

| Pin # | Single | Description | Remark |
|---|---|---|---|
| 1 | NC | No connection and do not connect with other NC | Keep Open |
| 2 | GDR | This pin is N-MOS gate control. | |
| 3 | RESE | Current sense input for control loop. | |
| 4 | NC | No connection and do not connect with other NC | Keep Open |
| 5 | VSH2 | Positive source voltage for Red | |
| 6 | TSCL | I2C clock for external temperature sensor | Connect to GND if not used |
| 7 | TSDA | I2C data for external temperature sensor | Connect to GND if not used |
| 8 | BS | Input interface setting. | Note 6-4 |
| 9 | BUSY_N | This pin indicates the driver status. | |
| 10 | RST_N | Global reset pin | Note 6-3 |
| 11 | DC | Serial communication Command/Data input | Note 6-2 |
| 12 | CSB | Serial communication chip select. | Note 6-1 |
| 13 | SCL | Serial communication clock input. | |
| 14 | SDA | Serial communication data input. | |
| 15 | VDDIO | IO voltage supply | |
| 16 | VCC | Digital/Analog power. | |
| 17 | VSS | Digital ground | |
| 18 | VDD | 1.5V voltage input &output | |
| 19 | VPP | OTP program power (10V) | |
| 20 | VSH1 | Positive source voltage | |
| 21 | VGH | Positive gate voltage | |
| 22 | VSL | Negative source voltage. | |
| 23 | VGL | Negative gate voltage. | |
| 24 | VCOM | VCOM driving voltage | |

1. Note 6-1: This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled for MCU communication: only when CSB is pulled LOW.

2. Note 6-2: This pin (DC) is Data/Command control pin connecting to the MCU.

When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled LOW, the data will be interpreted as command.

3. Note 6-3: This pin (RST_N) is reset signal input. The Reset is active low.

4. Note 6-4: This pin (BS) is for 3-line SPI or 4-line SPI selection. When it is “Low”, 4-line SPI is selected. When it is “High”, 3-line SPI (9 bits SPI) is selected.

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# 7. Reference Circuit

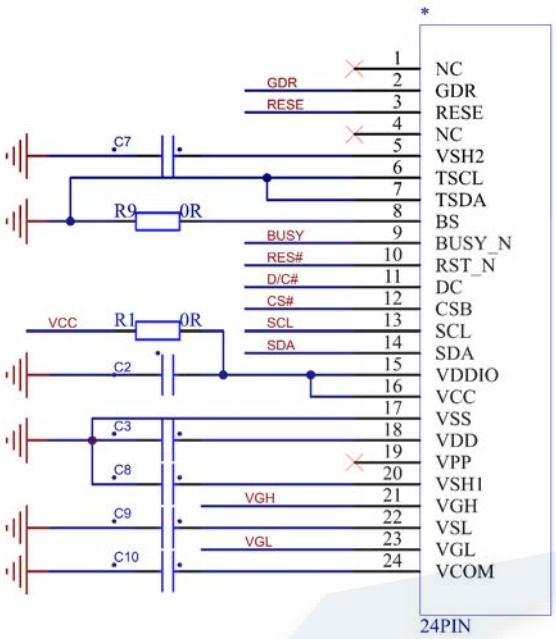



**Figure. 7-1**

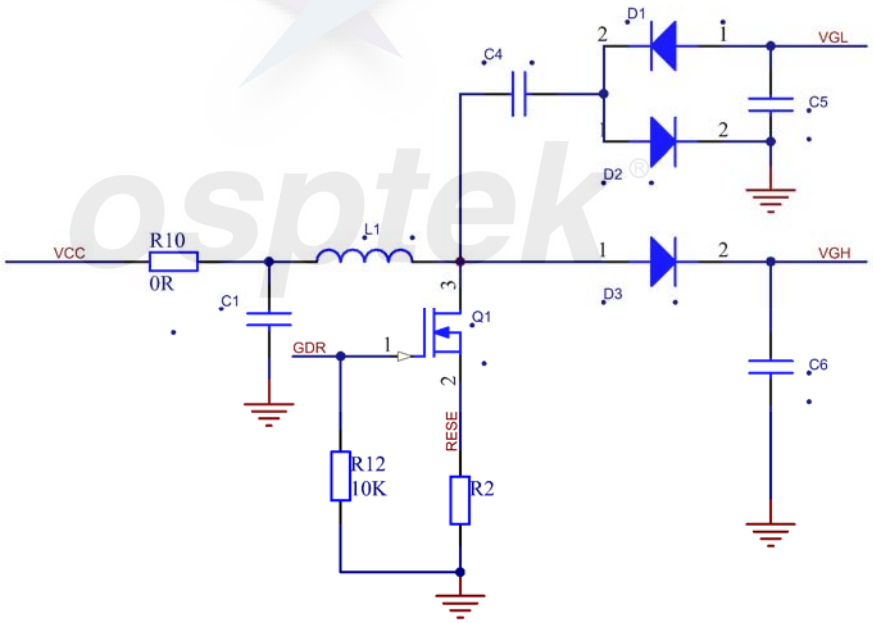



**Figure. 7-2**

| **Part Name** | **Value/Type** | SSD1681Value /requirement/Reference Part |
|---|---|---|
| C0-C1 | 1uF | X5R/X7R; Voltage Rating : 6V or 25V |
| C2-C7 | 1uF | 0402/0603/0805; X5R/X7R; Voltage Rating : 25V |
| C8 | 1uF | 0402/0603/0805; X5R/X7R; Voltage Rating : 25V |
| R1 | 2.2 ohm | 0402/0603/0805; 1% variation, 0.05W |
| D1-D3 | Diode | MBR0530<br>1) Reverse DC voltage 30V<br>2) Io:500mA<br>3) Forward voltage ≤ 430mV |
| Q1 | NMOS | Si1304BDL/NX3008NBK<br>1） Drain-Source breakdown Voltage=30V（min）<br>2） Vgs(th)=0.9V(Typ),1.3V(Max)<br>3） Rds on ≤2.1Ω & Vgs=2.5V |
| L1 | 47uH | CDRH2D18/LDNP-470NC<br>Io=500(Max) |
| CON24Pin | 0.5mm ZIF Socket | 24Pins,0.5mm pitch |

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## 8. ABSOLUTE MAXIMUM RATING

| Parameter | Symbol | Rating | Unit | Humidity | Unit | Note |
|---|---|---|---|---|---|---|
| Logic Supply voltage | VCC,VDDIO | -0.5~6.0 | V | 45~70 | %RH | Note 8-1 |
| Operation temperature range | TOPR | 0~50 | ℃ | 45~70 | %RH | |
| Storage temperature range | TSTG | -25~60 | ℃ | 45~70 | %RH | Note 8-2 |
| Transportation temperature range | TTTG | -25~60 | ℃ | 45~70 | %RH | Note 8-3 |
| GND | Ground | - | - | | - | Connect to Ground |

Note 8-1: Maximum ratings are those values beyond which damages to the device may occur. VCC: Digital power, VDDIO: IO power. Functional operation should be restricted to the limits in the Electrical Characteristics chapter.

Note 8-2: The display effect may be affected if the product is stored for more than 10days under the environment of above 60 degree. The recommended long storage temperature is 10℃~30℃.

Note 8-3: TTTG is the transportation condition, the transport time is within 10 days for -25℃~0℃/40℃~60℃ or within 5 days for 60℃~70℃

Note 8-4: The single pixel effect under the condition of above 35 degree cannot be guaranteed. The product can operate under the temperature -15~0℃ or 50~65℃, but the display effect cannot be guaranteed.

## 9. DC Characteristics

The following specifications apply for: VSS=0V, VCC=3.0V, TOPR=25±2℃.

| Symbol | Parameter | Test Condition | Applicable pin | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|---|
| VCC | VCC operation voltage | - | VCC | 2.4 | 3.0 | 3.6 | V |
| VIH | High level input voltage | - | SDA, SCL, CS#, D/C#, RES#, BS1 | 0.8VDDIO | - | - | V |
| VIL | Low level input voltage | - | | - | - | 0.2VDDIO | V |
| VOH | High level output voltage | IOH = -100uA | BUSY, | 0.9VDDIO | - | - | V |
| VOL | Low level output voltage | IOL = 100uA | | - | - | 0.1VDDIO | V |
| Iupdate | Module peak current | - | - | - | - | 40 | mA |
| Isleep | Deep sleep mode | VCC=3.0V | - | - | | 3 | uA |

- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by XingTai.
- VCOM value will be OTP before in factory.

## 10. Power Consumption

| Parameter | Symbol | Conditions | TYP | Max | Unit | Remark |
|---|---|---|---|---|---|---|
| Panel power consumption during update | - | 25℃ | | 80 | mAs | - |

MAs=update Average current ×update time

The Typical power consumption is measured using associated 25℃waveform with following pattern:




# 11. Optical characteristics

## 11.1 Optical Measurement Conditions

| Item | Symbol | Value | Unit | Note |
|---|---|---|---|---|
| Ambient Temperature | Ta | 25±2 | ℃ | Indoor testing |
| Ambient Humidity | Ha | 50±5 | %RH | - |
| Supply Voltage | VCC, VDDIO | 3.0 | V | - |
| illuminance | - | 800~1300 | Lux | |

Note 11-1: Image is updated with above condition

## 11.2 Optical Measurement

WS: White state, BS: Black state

| SYMBOL | PARAMETER | CONDITIONS | MIN | TYP. | MAX | UNIT | Note |
|---|---|---|---|---|---|---|---|
| △E | ghosting | - | - | - | 2 | | - |
| BS | Black State L* value | | - | - | 24 | | Note 11-2 |
| WS | White State L* value | | 66 | - | - | | Note 11-2 |

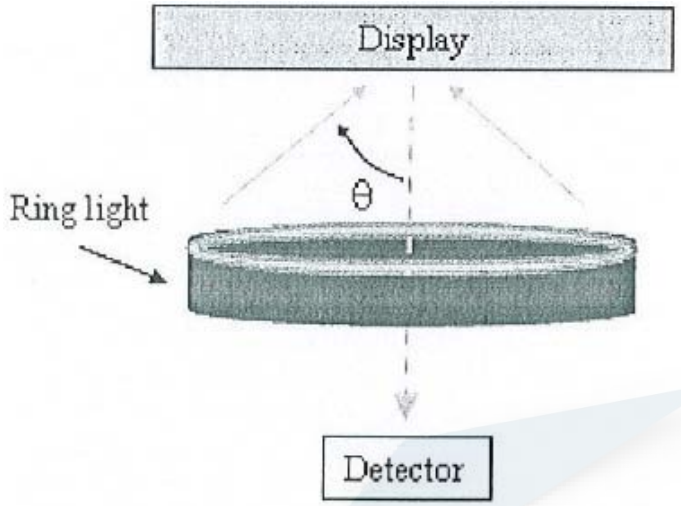
Note 11-2: Luminance meter: Eye - One Pro Spectrophotometer

Note 11-3:We don't guarantee 5 years pixels display quality for humidity below 45%RH or above 70%RH； Suggest Updated once a day;

## 11.3 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (Rl) and the reflectance in a dark area (Rd)：R1: white reflectance Rd: dark reflectance

CR = Rl/Rd

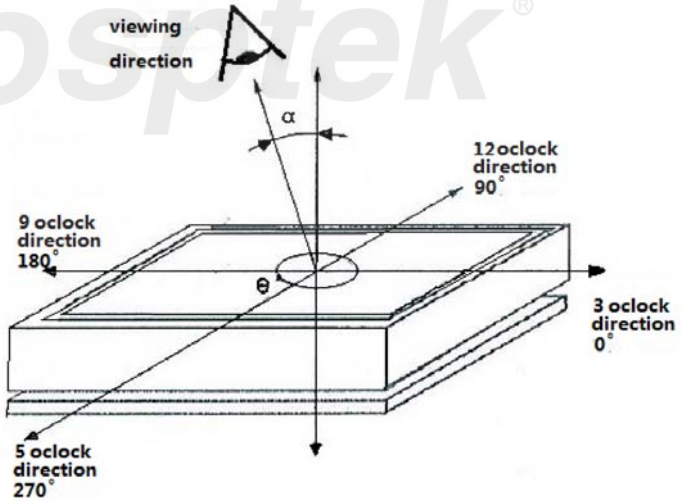



## 11.4 Reflection Ratio

The reflection ratio is expressed as:

R = Reflectance Factor white board x (L center / L white board)

L center is the luminance measured at center in a white area (R=G =B=1). L white board is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.



## 12. Handling Safety and Environmental Requirements

| WARNING |
| --- |
| The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap. |

| CAUTION |
| --- |
| The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. |
| Disassembling the display module can cause permanent damage and invalidate the warranty agreements. |
| IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed. |
| Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricity and other rough environmental conditions. |

| Mounting Precautions |
| --- |
| (1) It`s recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module. |
| (2) It`s recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force. |
| (3) You should adopt radiation structure to satisfy the temperature specification. |
| (4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction. |
| (5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS) |
| (6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS. |
| (7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading. |

| Data sheet status |
| --- |

| | |
|---|---|
| Product specification | The data sheet contains final product specifications. |

| **Limiting values** |
|---|
| Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability. |
| **Application information** |
| Where application information is given, it is advisory and dose not form part of the specification. |

| **Product Environmental certification** |
|---|
| ROHS |
| **REMARK** |
| All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation. |

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# 13. Reliability test

## 13.1 Reliability test items

| | TEST | CONDITION | METHOD | REMARK |
|---|---|---|---|---|
| 1 | High-Temperature Operation | T=40℃，RH=35%RH，For 240Hr | | |
| 2 | Low-Temperature Operation | T = 0℃ for 240 hrs | | |
| 3 | High-Temperature Storage | T=60℃ RH=35%RH For 240Hr<br>Test in white pattern | | |
| 4 | Low-Temperature Storage | T = -25℃ for 240 hrs<br>Test in white pattern | | |
| 5 | High Temperature, High-Humidity Operation | T=40℃，RH=90%RH，For 168Hr | | |
| 6 | High Temperature, High-Humidity Storage | T=60℃，RH=80%RH，For 240Hr<br>Test in white pattern | | |
| 7 | Temperature Cycle | -25℃(30min)~70℃(30min)，50 Cycle<br>Test in white pattern | | |
| 8 | Package Vibration | 1.04G,Frequency : 10~500Hz<br>Direction : X,Y,Z<br>Duration:1hours in each direction | Full packed for shipment | |
| 9 | Package Drop Impact | Drop from height of 100 cm on Concrete surface<br>Drop sequence:1 corner, 3edges, 6face<br>One drop for each. | Full packed for shipment | |
| 10 | UV exposure Resistance | 765 W/㎡ for 168hrs,40℃ | | |
| 11 | ESD test | Operation/Non-operation, contact mode ±2KV/Air mode ±6KV | | |

Actual EMC level to be measured on customer application.

Note1：Stay white pattern for storage and non-operation test.

Note2：Power off duration time is 30s

Note3：Continue testing after 2 hours at 20C~25C℃

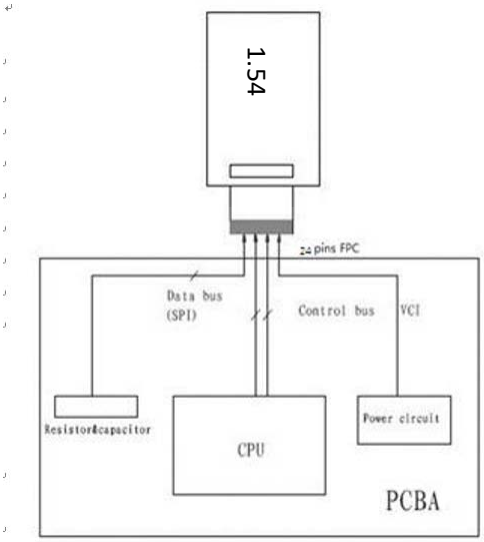
## 13.2 Product life time

The EPD Module is designed for a 5-year life-time with 25 ℃/60%RH operation assumption. Reliability estimation testing with accelerated life-time theory would be demonstrated to provide confidence of EPD lifetime.

## 13.3 Product warranty

Warranty conditions have to be negotiated between Xingtai and individual customers. Xingtai provides 12+1(one month delivery time) months warranty for all products which are purchased from Xingtai.

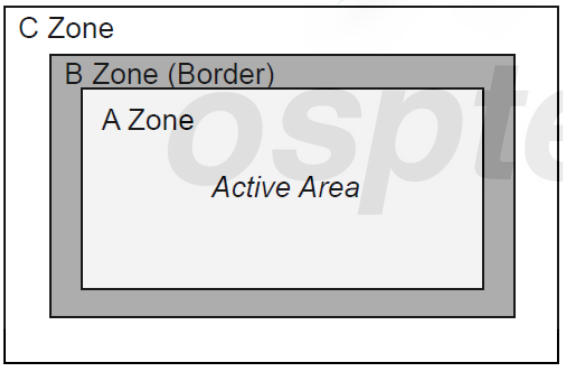
# 14. Block Diagram



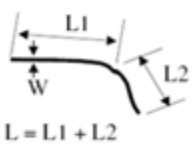

# 15. Shipment inspection specification

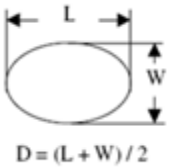
## 15.1 Zone Definition

A Zone: Active Area
B Zone: Border Area
C Zone: From B Zone edge to panel edge



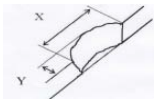
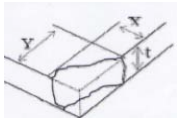

## 15.2 Line/Spot defect size



## 15.3 Point and line standard

| Shipment Inspection Standard | | | | | | |
|---|---|---|---|---|---|---|
| Equipment：Electrical test fixture, Point gauge | | | | | | |
| **Outline dimension** | 37.32(H)×31.80(V) ×0.86(D) | **Unit：** ㎜ | **Part-A** | Active area | **Part-B** | Border area |
| | | | | | | |
| **Environment** | **Temperature** | **Humidity** | **illuminance** | **Distance** | **Time** | **Angle** |
| | 19℃～25℃ | 50±5%RH | 800~1300Lux | 300 ㎜ | 35Sec | 45° |
| | | | | | | |
| **Defect type** | **Inspection method** | **Standard** | | **Part-A** | | **Part-B** |
| Spot defect | Visual/Film card （below 4.2inch） | D≤0.25 ㎜ | | Ignore | | Ignore |
| | | 0.25 ㎜＜D≤0.45 ㎜ | | N≤4 | | Ignore |
| | | D＞0.45 ㎜ | | Not Allow | | Ignore |
| | Visual/Film card （4.2~7.5inch） | D≤0.25 ㎜ | | Ignore | | Ignore |
| | | 0.25 ㎜＜D≤0.45 ㎜ | | N≤4 | | Ignore |
| | | 0.45 ㎜＜D≤0.6 ㎜ | | N≤2 | | Ignore |
| | | D＞0.6 ㎜ | | Not Allow | | Ignore |
| | Visual/Film card （above 7.5inch） | D≤0.45 ㎜ | | Ignore | | Ignore |
| | | 0.45 ㎜＜D≤0.6 ㎜ | | N≤4 | | Ignore |
| | | 0.6 ㎜＜D≤0.8 ㎜ | | N≤2 | | Ignore |
| | | D＞0.8 ㎜ | | Not Allow | | Ignore |
| PS/bubble | Visual/Film card （below 4.2inch） | D≤0.2mm | | Ignore | | Ignore |
| | | 0.2<D≤0.35mm | | N≤4 | | Ignore |
| | | D>0.35mm | | Not Allow | | Ignore |
| | Visual/Film card （above 4.2inch） | D≤0.4mm | | Ignore | | Ignore |
| | | 0.4<D≤0.6mm; DS>10mm | | N≤4 | | Ignore |
| | | D>0.6mm | | Not Allow | | Ignore |
| PS/Dirty | Visual/Film card | L≤1 ㎜,W≤0.03 ㎜ | | Ignore | | Ignore |
| | | L≤4 ㎜,0.03<W≤0.25 ㎜ | | N≤4 | | Ignore |
| | | W>0.25mm; L>4mm | | Not Allow | | Ignore |
| Display | Electric Display | Not Allow | | Not Allow | | Ignore |

| | | | | |
|---|---|---|---|---|
| malfunction | | | | |
| Display error | Electric Display | Not Allow | Not Allow | Ignore |
| TFT Scratch or line defect(include dirt) | Visual/Film card | L≤2 ㎜,W≤0.2 ㎜ | Ignore | Ignore |
| | | 2.0mm＜L≤9.0mm,0.2＜W≤0.3mm, | N≤4 | Ignore |
| | | L＞9 ㎜,W＞0.3 ㎜ | Not Allow | Ignore |
| TFT Corner /Edge chipping | Visual/Film card | X≤8mm,Y≤1mm, Do not affect the electrode circuit (Edge chipping) Ignore | | |
| | | X≤1mm,Y≤1mm, Do not affect the electrode circuit( (Corner chipping) Ignore | | |
| | |   | | |
| TFT warping | For 1.54~7.5inch, T≤2mm; For above 7.5inch, T≤3mm   | | | |
| Remark | 1.Cannot be defect & failure cause by appearance defect; | | | |
| | 2.Cannot be larger size cause by appearance defect; | | | |
| | L=long W=wide D=point size N=Defects NO | | | |

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## 15.4 Barcode



FF

ABBBBBBBBB

CCCDDDEEEE

What is the QR code after picture being scanned and read? It consist of 30 numbers and letters, here is explanation for it.

QR Code: ABBBBBBBBBFFCCCDDDEEEEGHIJKKLL

① A means Factory code, e.g.: H (HINK)
② BBBBBBBBB refers to product model name, which is in consistent with specification part number. E.g.: E0213A189 or E0266A120; when the total code is less than 9, it would be added suffix with **. e.g.: E027A45** or E0154A78*
③ FF means production workshop
④ CCC means date of production
⑤ DDD means production batch
⑥ EEEE means FPL lot
⑦ G means MP/Sample/Trial/Repair
⑧ HIJ:(H——TFT;I——PS;J——EC Glue)
⑨ KK means IC type
⑩ LL means serial number

## 16. Packing

Full carton: 12 pcs tray with products and 1 pcs empty tray.
Last carton: less 12pcs tray with products and 1 pcs empty tray. The packager will add the right amount of EPE to box so that the added EPE's height approximately close to the EPE height around the inside of the box.

